

RISHABH GOEL
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Curious and driven technologist who blends math, code, and creativity to explore complex ideas. Passionate about scientific discovery, mentoring peers, and making hard concepts approachable.

EDUCATION

Rock Ridge High School, Ashburn VA — High School Diploma | Sep 2020 – Jun 2025
Relevant coursework: AP Physics C: Mechanics, AP Calculus BC, AP Statistics, AP Computer Science A

University of Maryland, College Park MD — Physics Major | Sep 2025 – May 2029
Relevant coursework: Introductory Physics: Fields • Experimental Physics I: Mechanics and Waves • Multivariable Calculus, Linear Algebra and Differential Equations I (Honors) • Object Oriented Programming I and II (exemption exam)

EXPERIENCE & PROJECTS

ZeroClone — Open-Source Creator | Apr 2025 – Present
- Building a general learning algorithm for two-player perfect information games, inspired by GDM's AlphaZero.
- Implemented batched-GPU self-play pipeline and plug-and-play game adapters (C++/Python).

UMD LHCb Group — Undergraduate Researcher | Dec 2025 – Present
- Using bootstrapping methods to estimate uncertainty in one of the trigger mechanisms in the CERN LHC.
- Collaborated with grad students, worked with large datasets using ROOT and similar tools.

Rock Ridge HS Competitive Programming Club — President | Sep 2021 – Jun 2025
- Led and coached 15 members; placed Top-10 in multiple Chesapeake-region university contests.
- Organized weekly problem-solving workshops and collectively solved 30+ 1800 rated problems on Codeforces.

Yogic Transformers FTC Robotics Team — Autonomous Algorithms Lead | Sep 2021 – Jun 2023
- Developed sensor-fusion & vision pipelines for fully autonomous navigation; qualified for state semi-finals twice.

Quantiphi — Machine Learning Engineer Intern | Jun 2020 – Jul 2020
- Prototyped a computer-vision research project under senior mentorship and presented findings to leadership.

SKILLS

Programming Languages: Python • C++ • Java • Bash/Zsh • Embedded C/C++ (Arduino)
AI / ML & Data: PyTorch • Reinforcement Learning (MCTS, self-play) • NumPy/Pandas
Systems & DevOps: Linux admin • Docker • CMake/build pipelines • Git/GitHub
Math & Science: Mechanics • Electromagnetism • Probability/Statistics • Calculus
Agentic Coding: Claude Code • Codex

INTERESTS: particle physics, cosmology, deep learning, character-driven literature ("The Goldfinch," "A Little Life", "Pachinko") and cinema ("Whiplash," "The Pianist", "Shawshank Redemption").